BREAST CARE-MA DIAGNOSTIC DECISION SUPPORT REPORT

Department of Information Technology | Nehru Institute of Engineering and Technology
Explainable Multi-Agent AI System for Ultrasound Modality

Case ID:	BUSI-MALIGNANT-8827	**Date & Time:**	2026-10-31 00:13:21
Patient Age:	48	**Family History:**	Yes
Breast Density:	C - Heterogeneously dense	**Symptoms:**	Palpable nodule

1. Quantitative AI Findings & Model Outputs

Parameter	Result	Clinical Meaning
AI Classification	**MALIGNANT**	Calibrated Malignant Probability: 40.6%
Tumor Detection	Lesion Detected	Relative Area: 40.50%
Morphology (Agent 3)	Irregular / Indistinct	Aspect Ratio: 1.33, Solidity: 0.74
Acoustic Shadowing	Absent	Acoustic attenuation distal to mass
ACR BI-RADS Category	**BI-RADS 4C**	**High Suspicion (50% - 85%)**

2. Retrieved Evidence & Clinical Guidelines (Agent 4)

• **Moderate Suspicion of Malignancy (>10% to <=50%)** (ACR BI-RADS Atlas 5th Edition / NCCN Guidelines): Solid lesion with indistinct margins or non-parallel orientation without definite spiculation. Malignancy risk spans 10% to 50%.

• **High Suspicion of Malignancy (>50% to <95%)** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): Irregular hypoechoic mass with angular margins and posterior acoustic shadowing, highly concerning for invasive ductal carcinoma (IDC).

• **Classic Malignant Presentation (>=95% Probability)** (ACR BI-RADS Atlas 5th Edition / Society of Breast Imaging): Classic presentation: Spiculated margins, taller-than-wide (non-parallel) orientation, hypoechoic with acoustic shadowing, microcalcifications. Malignancy risk exceeds 95%.

3. Recommended Clinical Follow-up Actions

• Urgent ultrasound-guided core needle biopsy recommended.

• Immediate multidisciplinary breast oncology and surgical consultation.

• Consider contrast-enhanced breast MRI for staging if invasive disease is confirmed.

REGULATORY & SAFETY DISCLAIMER: NOTICE: Breast CARE-MA is an AI decision-support research prototype developed for academic evaluation. It does not provide definitive medical diagnosis and must never supersede clinical judgment by a licensed radiologist or oncologist.